

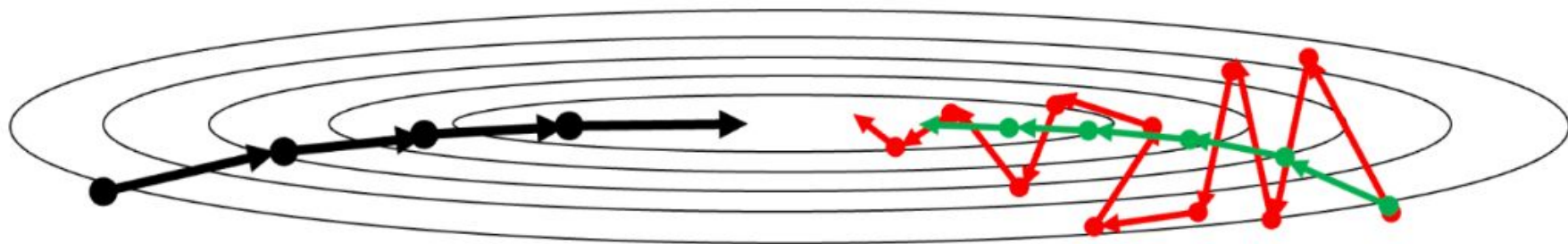
# Introduction to Neural Networks

Seminar - Week 5 - Optimizers

# Assignment 2 Goals

- Implement optimizers
  - SGD with Momentum
  - RMSProp
  - Adam

# Gradient Descent - Momentum



Gradient Descent

$$W := W - \alpha dW$$

$$b := b - \alpha db$$

Gradient Descent with Momentum

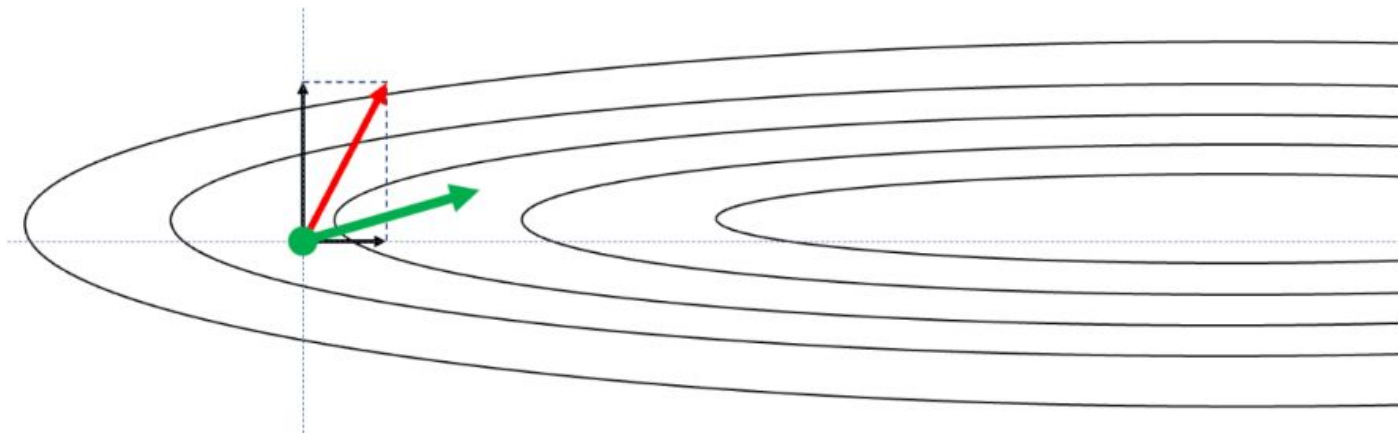
$$V_{dW} := (1 - \beta)dW + \beta V_{dW}$$

$$V_{db} := (1 - \beta)db + \beta V_{db}$$

$$W := W - \alpha V_{dW}$$

$$b := b - \alpha V_{db}$$

# RMSProp



$$S_{dW} := (1 - \beta)dW^2 + \beta S_{dW}$$

$$S_{db} := (1 - \beta)db^2 + \beta S_{db}$$

$$W := W - \alpha \frac{dW}{\sqrt{S_{dW}}}$$

$$b := b - \alpha \frac{db}{\sqrt{S_{db}}}$$

# Adam

$$V_{dW} := (1 - \beta_1)dW + \beta_1 V_{dW}$$

$$S_{dW} := (1 - \beta_2)dW^2 + \beta_2 S_{dW}$$

$$\widehat{V}_{dW} := \frac{V_{dW}}{1 - \beta_1^t}$$

$$\widehat{S}_{dW} := \frac{S_{dW}}{1 - \beta_2^t}$$

$$W := W - \alpha \frac{\widehat{V}_{dW}}{\sqrt{\widehat{S}_{dW} + \epsilon}}$$

Small value  
for numerical stability

# Week 5 - TASK

- Implement optimizers
  - SGD with Momentum
  - RMSProp
  - Adam

[https://github.com/vgg-fiit/zneus\\_2025](https://github.com/vgg-fiit/zneus_2025)

# Project 1 (9b)

- MLP
  - Data analysis
  - Data preprocessing and normalization
  - Data split
  - Configuration
  - Experiment tracking
  - Experiments - meaningful based on the results of previous experiments
    - Including improvement techniques (e.g. Dropout, Normalization layers, Skip Connections, Bottleneck Layers, ...)
  - Hyperparameter search (grid/random) / sweep
  - Results and evaluation metrics
  - Clear code
  - Markdown documentation and comments
  - Final presentation of projects
  - Effort on consultations

# Project 1

- Consultation 1 - minimal requirements
  - Select team ([google sheet](#))
  - Select task (maximum 2 teams per task, first come, first serve)
  - Download and examine dataset
  - Exploratory data analysis ([example](#))
  - Select evaluation metrics
  - Check [THIS](#) and [THIS](#) video



# Project 1 - List of topics

## Credit Risk Assessment

- Binary Classification
- Dataset - <https://api.openml.org/d/31>

## Speed Dating

- Binary Classification
- Dataset - <https://api.openml.org/d/40536>

## Diabetes 130 US

- Binary Classification
- Dataset - <https://api.openml.org/d/45069>

## Bioresponse

- Binary Classification
- Dataset - <https://api.openml.org/d/4134>

## Magic Telescope

- Binary Classification
- Dataset - <https://www.openml.org/data/download/54003/MagicTelescope.arff>  
<https://archive.ics.uci.edu/dataset/159/magic+gamma+telescope>
- paper - <https://publikationen.bibliothek.kit.edu/270043064/3813660>
  - paper about the creation of the data

## Detroit

- Regression
- Dataset - <https://api.openml.org/d/552>

## Houses

- Regression
- Dataset - <https://api.openml.org/d/537>

## Covid 19

- Multiclass Classification
- Dataset - <https://www.kaggle.com/datasets/meirnazri/covid19-dataset>

## Steel Plates Fault

- Multiclass Classification/Binary classification
- Bonus point - Utilize both approaches (either compare, or use together)
- Dataset - <https://api.openml.org/d/1504>